

CN Unit 2 Previous Year Questions

Computer Networks (CN) · Unit 2 · Data Link Layer & MAC · 2024

Questions collected from previous year university examinations (2020–2024).

Part A — Short answer (2 marks)

1. Compare Framing with Sliding Window Protocols. When would you prefer one over the other? [2020]
2. Discuss the role of Error Detection in Data Link Layer & MAC. [2023]
3. Define Hamming Code. Derive or justify the key properties involved. [2022]
4. Discuss the role of Sliding Window Protocols in Data Link Layer & MAC. [2021]

Part B — Long answer (10 marks)

1. Write short notes on ALOHA and CSMA/CD. [2021]
2. Define Ethernet and Switching. Derive or justify the key properties involved. [2022]
3. Define Framing. Derive or justify the key properties involved. [2023]
4. Illustrate Error Detection with a worked example. [2023]
5. What is meant by Hamming Code? Discuss its advantages and limitations. [2024]
6. What is meant by Sliding Window Protocols? Discuss its advantages and limitations. [2021]
7. Discuss the role of ALOHA and CSMA/CD in Data Link Layer & MAC. [2021]
8. Illustrate Ethernet and Switching with a worked example. [2022]
9. Discuss the role of Framing in Data Link Layer & MAC. [2023]
10. Discuss the role of Error Detection in Data Link Layer & MAC. [2023]